Supplementary material

Dilution-dependent antibacterial activity of honey: a systematic review and quantitative synthesis of physicochemical and microbiological evidence

For submission to International Journal of Food Microbiology.

S1. Search strategy and yields

All searches were executed programmatically on the dates shown; raw API responses are retained in the repository archive. Europe PMC queries were capped at 500 retrieved records per query and Crossref at 100, ranked by relevance.

Table S1. Full search queries, dates and yields.

Database	Query	Date	Hits	Retrieved
pubmed	honey[Title/Abstract] AND (antibacterial[Title/Abstract] OR antimicrobial[Title/Abstract]) AND ("minimum inhibitory"[Title/Abstract] OR dilution[Title/Abstract])	2026-09-26	237	237
pubmed	honey[Title/Abstract] AND (bacteriostatic[Title/Abstract] OR bactericidal[Title/Abstract]) AND (dilution[Title/Abstract] OR concentration[Title/Abstract])	2026-09-26	104	104
pubmed	honey[Title/Abstract] AND ("hydrogen peroxide"[Title/Abstract] OR "glucose oxidase"[Title/Abstract] OR peroxide[Title/Abstract] OR catalase[Title/Abstract]) AND (bacteria*[Title/Abstract] OR	2026-09-26	178	178

	antimicrobial[Title/Abstract])			
pubmed	honey[Title/Abstract] AND ("water activity"[Title/Abstract] OR osmotic[Title/Abstract] OR osmolarity[Title/Abstract] OR "artificial honey"[Title/Abstract] OR "sugar solution"[Title/Abstract]) AND (antibacterial[Title/Abstract] OR antimicrobial[Title/Abstract] OR inhibitory[Title/Abstract])	2026-09-26	82	82
pubmed	honey[Title/Abstract] AND (methylglyoxal[Title/Abstract] OR MGO[Title/Abstract] OR "non-peroxide"[Title/Abstract] OR "nonperoxide"[Title/Abstract]) AND (antibacterial[Title/Abstract] OR antimicrobial[Title/Abstract] OR bactericidal[Title/Abstract])	2026-09-26	147	147
pubmed	honey[Title/Abstract] AND ("inhibition zone"[Title/Abstract] OR "zone of inhibition"[Title/Abstract] OR "well diffusion"[Title/Abstract] OR "agar dilution"[Title/Abstract] OR "broth dilution"[Title/Abstract])	2026-09-26	160	160
europepmc	honey AND (antibacterial OR antimicrobial OR bacteriostatic)	2026-09-26	10833	500

	AND (dilution OR concentration OR "minimum inhibitory" OR MIC) IN_EPMC:y OPEN_ACCESS:y			
europepmc	honey AND ("hydrogen peroxide" OR "glucose oxidase" OR catalase) AND (antibacterial OR antimicrobial)	2026-09-26	4165	500
europepmc	honey AND ("water activity" OR osmotic OR "artificial honey" OR "sugar solution") AND (antibacterial OR antimicrobial OR inhibitory)	2026-09-26	2592	500
europepmc	honey AND (methylglyoxal OR MGO OR "non-peroxide") AND (antibacterial OR antimicrobial OR bactericidal)	2026-09-26	994	500
europepmc	honey AND ("inhibition zone" OR "zone of inhibition" OR "well diffusion" OR "agar dilution" OR "broth microdilution") AND (MIC OR antibacterial OR antimicrobial)	2026-09-26	2048	500
crossref	honey antibacterial dilution minimum inhibitory concentration	2026-09-26	658913	100
crossref	honey hydrogen peroxide antibacterial activity dilution	2026-09-26	2095135	100
crossref	honey water activity osmotic antibacterial artificial honey	2026-09-26	4340473	100

crossref	manuka honey methylglyoxal antibacterial non-peroxide	2026-09-26	1997456	100
crossref	honey inhibition zone antibacterial concentration	2026-09-26	1034389	100
pubmed_citemine	Blair SE[Author] AND honey; Irish J[Author] AND honey AND antibacterial; Roberts AEL[Author] AND honey; Henriques AF[Author] AND honey; Maddocks SE[Author] AND honey; Allen KL[Author] AND honey AND antibacterial; Molan PC[Author] AND honey; Voidarou C[Author] AND honey; Cooper RA[Author] AND honey AND antibacterial; Carnwath R[Author] AND honey; Alnaqdy[Author] AND honey; Lusby PE[Author] AND honey; Jenkins R[Author] AND honey AND antibacterial; Wilkinson JM[Author] AND honey; Brudzynski K[Author] AND honey AND antibacterial; Bucekova M[Author] AND honey; Kwakman PHS[Author] AND honey; Johnston M[Author] AND	2026-09-26	134	134

	honey AND antibacterial; Girma A[Author] AND honey; Masoura M[Author] AND honey; Grecka K[Author] AND honey; Nolan VC[Author] AND honey; Oryan A[Author] AND honey AND antibacterial; Cokcetin NN[Author] AND honey; Bouacha M[Author] AND honey; Combarros-Fuertes P[Author] AND honey; Shepherd P[Author] AND honey; Carter DA[Author] AND honey			

S2. Eligibility decision rules

A deterministic rule-based classifier assigned each screened record to include / exclude / review_flag using title-and-abstract patterns. The complete rules are implemented in scripts/04_eligibility.py in the released code. Decision counts:

Table S2. Eligibility decisions by rule outcome.

Decision rule outcome	n records
meets criteria	381
no quantitative antibacterial outcome	175
formulation with inseparable matrix	83
combination therapy - verify honey-alone arm	23
non-bacterial outcome only	21
non-honey bee product is main article	19
clinical/in-vivo without microbiological assay	13
post-hoc audit: not honey-specific primary data	8
viral/phage outcome only	3

S3. Extraction and provenance

Each extracted row carries study_id, source table identifier, row and column position, raw cell text, normalized outcome, unit, unit basis and an auto/manual flag. Manually curated mechanistic rows (n = 0) take precedence over automatically extracted rows for the same cell. Outcome categories in the harmonized dataset:

Table S3. Harmonized outcome categories.

Outcome	n rows
MIC	2572
zone	1433
MBC	710
H2O2	94
series	84
phenol_equiv	69
TPC	49
GOX	33
aw	19
n_outperform_manuka	19
phenol	17
pct_viable	15
color	12
MBEC	12
MIC50	10
pH	10
pct_inhib_biofilm	8
MGO	4
PLA	4

Reported concentration bases were preserved and analyzed separately:

Table S4. Concentration unit bases (not pooled).

Unit basis	n rows
% (basis unspecified)	1396
mm	1386
other	781
%v/v	683
%w/v	514
mass	246
%w/w	99
%w/v phenol	69

S4. Extraction-selection characterization

Of 404 eligible studies, 247 had a resolvable PubMed Central identifier, 226 had an open-access JATS XML full text, and 102 yielded machine-extractable outcome data. Extracted studies were more recent (median year 2022 vs. 2020; Mann-Whitney p = 2.04e-04). The most frequent journals among non-extracted eligible studies:

Table S5. Journals most represented among non-extracted eligible studies.

Journal (non-extracted eligible)	n studies
Antibiotics (Basel, Switzerland)	14
Foods (Basel, Switzerland)	11
Food chemistry	10

Journal of applied microbiology	9
Journal of medicinal food	8

Reasons for non-extraction fall into three classes: (i) no resolvable PMC identifier (typically journals not deposited in PMC or closed-access publisher versions); (ii) PMC identifier present but no open-access full text; (iii) full text retrieved but containing no outcome table matching the documented parsing layouts (data reported only in figures, in prose, or in non-tabular supplements). Counts are reconciled in Fig. 1 of the main text.

S5. Reporting-quality / risk-of-bias framework

Five domains, each scored 0-2: D1 assay standardization; D2 comparator quality; D3 dispersion reporting; D5 mechanistic perturbation; D6 reporting completeness. Concentration-series availability is an availability flag, not a quality score. Ratings: low (total >= 8), some (5-7), high (<= 4).

Table S6. Per-study reporting-quality scores.

study_id	D1_assay_std	D2_comparator	D3_dispersion	conc_series_available	D5_mechanism	D6_reporting	total	rating
PMC10000891	1	1	2	0	1	1	6	some
PMC10141347	1	1	1	0	2	1	6	some
PMC10211265	2	1	1	0	1	1	6	some
PMC10215129	2	1	1	0	1	2	7	some
PMC10386826	1	1	1	0	2	1	6	some
PMC10534861	2	1	1	0	1	1	6	some
PMC10617706	1	1	2	0	2	1	7	some
PMC10622673	2	1	1	0	1	1	6	some
PMC10856440	1	1	1	0	1	2	6	some
PMC11052219	2	1	1	0	1	1	6	some
PMC11068765	2	1	1	0	1	2	7	some
PMC11109849	1	1	1	0	1	2	6	some
PMC11111008	2	2	1	0	2	1	8	low
PMC11120828	1	1	1	0	1	2	6	some
PMC11279380	2	1	1	0	1	1	6	some
PMC11311187	1	1	1	0	1	2	6	some

PMC11376693	1	1	1	1	1	2	6	some
PMC11561629	2	1	1	0	1	1	6	some
PMC11596660	2	1	1	0	1	1	6	some
PMC11675739	2	1	1	0	1	1	6	some
PMC11682447	2	1	1	0	2	1	7	some
PMC11792292	1	1	1	1	1	1	5	high
PMC11799142	2	1	1	0	1	2	7	some
PMC11851655	2	1	1	0	1	1	6	some
PMC11858134	2	1	1	0	1	1	6	some
PMC11941359	2	1	1	0	1	2	7	some
PMC12110744	2	1	1	0	1	1	6	some
PMC12175512	2	1	1	0	2	1	7	some
PMC12189776	2	1	1	0	1	1	6	some
PMC12240352	1	1	1	0	1	2	6	some
PMC12249035	2	1	1	0	1	1	6	some
PMC12379519	1	1	1	0	1	2	6	some
PMC12532491	1	1	1	1	1	2	6	some
PMC12686235	2	1	1	0	1	1	6	some
PMC12751895	2	1	1	0	1	1	6	some
PMC12861962	1	1	1	0	1	2	6	some
PMC12929673	2	1	1	0	1	1	6	some
PMC12963514	2	1	1	0	1	1	6	some
PMC13035723	2	1	1	0	1	2	7	some
PMC13068764	2	1	1	0	1	1	6	some
PMC13072155	2	1	1	0	1	1	6	some
PMC13315830	2	1	2	0	2	1	8	low
PMC13	2	1	1	0	1	1	6	some

379289								
PMC13460517	2	2	1	0	1	1	7	some
PMC13514261	1	1	1	0	1	2	6	some
PMC2478674	2	1	1	1	1	1	6	some
PMC2942791	2	1	1	0	1	2	7	some
PMC3042689	2	1	1	0	1	2	7	some
PMC3273858	1	1	2	0	2	1	7	some
PMC3389331	2	1	1	0	2	1	7	some
PMC3406342	2	2	1	0	2	1	8	low
PMC3716576	2	1	1	0	2	2	8	low
PMC4023514	2	1	1	1	1	1	6	some
PMC4029012	2	1	1	0	1	1	6	some
PMC4206207	1	1	1	0	1	2	6	some
PMC4355501	1	1	2	0	1	1	6	some
PMC4398880	2	1	1	0	1	1	6	some
PMC4417269	2	1	1	0	1	2	7	some
PMC4751499	2	1	1	0	1	1	6	some
PMC4783026	1	1	1	0	1	1	5	high
PMC4919268	2	1	1	0	1	1	6	some
PMC5489923	2	1	1	0	1	2	7	some
PMC6027671	2	1	1	0	1	2	7	some
PMC6034916	1	1	1	0	1	2	6	some
PMC6252015	1	1	1	0	1	2	6	some
PMC6268398	2	1	1	0	1	1	6	some
PMC6269728	2	2	1	0	1	1	7	some
PMC6313337	2	1	1	0	1	2	7	some
PMC6332115	1	1	1	0	1	2	6	some

PMC6397887	2	1	1	0	1	2	7	some
PMC6425412	1	1	1	0	1	2	6	some
PMC6659206	2	2	2	1	2	2	10	low
PMC6814216	2	1	1	0	1	1	6	some
PMC6881396	2	1	1	0	2	1	7	some
PMC6977423	1	1	1	0	1	2	6	some
PMC7076972	1	2	1	0	1	2	7	some
PMC7142649	2	1	1	0	1	1	6	some
PMC7396681	2	1	1	0	1	2	7	some
PMC7451694	2	1	1	0	1	1	6	some
PMC7719625	2	2	1	0	1	1	7	some
PMC7884280	2	1	1	0	1	1	6	some
PMC7936102	2	1	1	0	2	2	8	low
PMC8021062	1	1	1	0	2	2	7	some
PMC8125924	1	1	1	0	2	1	6	some
PMC8209695	1	1	1	0	2	1	6	some
PMC8234392	2	2	1	1	2	2	9	low
PMC8272120	2	1	1	0	1	1	6	some
PMC8304482	1	2	1	0	1	2	7	some
PMC8307090	1	1	1	0	1	2	6	some
PMC8626341	2	1	1	0	2	1	7	some
PMC8868498	2	1	1	0	1	1	6	some
PMC8944737	2	1	1	1	2	2	8	low
PMC8997407	2	1	1	0	1	1	6	some
PMC9219755	2	1	1	0	1	1	6	some
PMC9333225	2	1	1	0	2	1	7	some
PMC93	2	1	1	0	1	2	7	some

71847								
PMC9415335	2	1	1	0	1	2	7	some
PMC9505312	2	1	1	0	1	2	7	some
PMC9607359	2	1	1	0	1	1	6	some
PMC9703798	2	1	1	0	1	2	7	some
PMC9906859	2	1	2	0	1	2	8	low
PMC9958606	1	1	1	0	1	1	5	high

S6. Extended analyses and sensitivity results

Spearman associations between honey-level physicochemical parameters and antibacterial outcomes (joined on study and normalized sample label):

Table S7. Chemistry-outcome Spearman correlations.

Parameter	Outcome	n	n studies	rho	p
H2O2	MIC	29	2	0.460	0.0122
MGO	MIC	116	1	-0.517	2.78e-09
aw	MIC	20	1	-0.058	0.807
pH	MIC	359	7	0.071	0.179
pH	zone	53	2	-0.518	6.99e-05
TPC	MIC	172	5	-0.152	0.0459
TPC	zone	54	1	0.259	0.0586
moisture	MIC	112	3	0.165	0.0812
moisture	zone	45	1	-0.068	0.658

Leave-one-study-out cross-validation of outcome models (log10(MIC)):

Table S8. Leave-one-study-out cross-validation (n = 155, 3 studies). Negative R2 indicates worse-than-null out-of-study performance.

Model	Predictors	CV R2	CV RMSE
Null	-	-0.119	0.231
Chemistry	pH, TPC	-32.317	1.261
Label only	manuka/medical-grade flag	-0.119	0.231
Combined	chemistry + label	-32.317	1.261

Sensitivity analyses:

Table S9. Sensitivity analyses on MIC subsets.

Subset	n rows	n studies	Median MIC (%)
Low reporting-	512	8	16.835

concern studies			
Manuka-labelled excluded	2172	-	18.0
Manuka-labelled only	146	-	15.0
% w/v basis only	369	-	15.0
% v/v basis only	481	-	12.5
Studies measuring water activity	126	2	25.0
Studies measuring H2O2	226	7	16.0
Peer-reviewed only	2318	-	17.5

Leave-one-study-out resampling of the manuka-vs-other MIC contrast returned non-significant results ($p > 0.05$) in 0% of iterations.

Natural-vs-artificial pairs by study (direction-normalized potency ratio; >1 indicates greater natural-honey activity):

Table S10. Natural-vs-artificial potency ratios by study.

Study	Outcome	n pairs	Median potency ratio
PMC11111008	MIC	3	1.19
PMC13460517	MIC	4	1.95
PMC3406342	MIC	1	1.30
PMC6659206	MIC	3	2.69
PMC7076972	zone	4	n/a
PMC8234392	MBC	2	1.06
PMC8234392	MIC	2	1.06
PMC8304482	zone	6	1.46

S7. Studies included in the quantitative synthesis

Full author-year references for the 102 studies from which quantitative data were extracted. PMCID identifiers are retained in the released dataset (data/extraction.csv) for row-level provenance.

Table S11. References of studies included in the quantitative synthesis (alphabetical).

#	Reference
1	Abdulraheem Al-Kafaween, M., Al-Jamal, H.A., Mohd Hilmi, A.B., Elsahoryi, N.A., Jaffar, N., Khairi Zahri, M. (2020). Antibacterial properties of selected Malaysian Tualang honey against Pseudomonas aeruginosa and Streptococcus pyogenes. Iranian Journal of Microbiology. https://doi.org/10.18502/ijm.v12i6.5031
2	Aburayyan, W.S., Seder, N., Al-fawares, O., Fararjeh, A., Majali, I.S., Al-Hajaya, Y. (2024). Characterization of Antibiofilm and Antimicrobial Effects of Trigona Stingless Bee Honey Compared to Stinging Bee Centaurea hyalolepis and Citrus Honeys. Journal of Evidence-Based Integrative

	Medicine, 29, 2515690X241271978. https://doi.org/10.1177/2515690x241271978
3	Adgaba, N., Al-Ghamdi, A., Sharma, D., Tadess, Y., Alghanem, S.M., Ali Khan, K., Javed Ansari, M., Mohamed, G.K.A. (2020). Physico-chemical, antioxidant and anti-microbial properties of some Ethiopian mono-floral honeys. Saudi Journal of Biological Sciences, 27(9), 2366-2372. https://doi.org/10.1016/j.sjbs.2020.05.031
4	Ailli, A., Zibouh, K., Eddamsyry, B., Drioiche, A., Fetjah, D., Ayyad, F.Z., Mothana, R.A., Hawwal, M.F., Radi, M., Tarik, R., Elomri, A., Mouradi, A., Zair, T. (2024). Physicochemical Characterization of Moroccan Honey Varieties from the Fez-Meknes Region and Their Antioxidant and Antibacterial Properties. Metabolites, 14(7), 364. https://doi.org/10.3390/metabo14070364
5	Akpınar, S., Mutlu, N. (2025). Multidimensional analysis of honey from Eastern Anatolia (Kars): Pollen spectrum, physicochemical properties, and antimicrobial activity. PLOS One, 20(7), e0327861. https://doi.org/10.1371/journal.pone.0327861
6	Allcott, G.J., Evans, J., Merry, T.L., Cox, J.A.G. (2026). Unique Mānuka Factor (UMF)-dependent antimicrobial activity of Mānuka honey against respiratory pathogens cannot be explained by sugar and methylglyoxal alone. Microbiology, 172(8). https://doi.org/10.1099/mic.0.001746
7	Alotibi, I.A., Harakeh, S.M., Al-Mamary, M., Mariod, A.A., Al-Jaouni, S.K., Al-Masaud, S., Alharbi, M.G., Al-Hindi, R.R. (2018). Floral markers and biological activity of Saudi honey. Saudi Journal of Biological Sciences, 25(7), 1369-1374. https://doi.org/10.1016/j.sjbs.2018.05.021
8	Alshaybawee, M.R., Asgari, S., Ghadersoltani, P., Mehrabian, A., Saniee, P. (2025). Exploring the antibacterial and anti-biofilm activity of two Iranian medical-grade kinds of honey on multidrug-resistant Pseudomonas aeruginosa. BMC Complementary Medicine and Therapies, 25(1), 39. https://doi.org/10.1186/s12906-025-04778-1
9	Alzahrani, H.A., Alsabehi, R., Boukraâ, L., Abdellah, F., Bellik, Y., Bakhotmah, B.A. (2012). Antibacterial and Antioxidant Potency of Floral Honeys from Different Botanical and Geographical Origins. Molecules, 17(9), 10540-10549. https://doi.org/10.3390/molecules170910540
10	Anand, S., Deighton, M., Livanos, G., Morrison, P.D., Pang, E.C.K., Mantri, N. (2019). Antimicrobial Activity of Agastache Honey and Characterization of Its Bioactive Compounds in Comparison With Important Commercial Honeys.

	Frontiers in Microbiology, 10, 263. https://doi.org/10.3389/fmicb.2019.00263
11	Assaggaf, H.M., Naceiri Mrabti, H., Rajab, B.S., Attar, A.A., Hamed, M., Sheikh, R.A., Omari, N.E., Menyiy, N.E., Belmehdi, O., Mahmud, S., Alshahrani, M.M., Park, M.N., Kim, B., Zengin, G., Bouyahya, A. (2022). Singular and Combined Effects of Essential Oil and Honey of Eucalyptus Globulus on Anti-Inflammatory, Antioxidant, Dermatoprotective, and Antimicrobial Properties: In Vitro and In Vivo Findings. Molecules, 27(16), 5121. https://doi.org/10.3390/molecules27165121
12	Ayub, R., Umer, M., Maan, A.A., Rasool, B., Khan, M.K.I., Younis, T., Abbas, S., Sajjad, M., Kaleem, I., Imran, M., Ullah, A., Afzal, M.S., Shah, Z.H., Ahmed, S., Aslam, F., Chaudhary, N., Afzal, M.I. (2020). Antibiotics, Acid and Heat Tolerance of Honey adapted Escherichia coli, Salmonella Typhi and Klebsiella pneumoniae. Foods, 9(3), 311. https://doi.org/10.3390/foods9030311
13	Balázs, V.L., Nagy-Radványi, L., Filep, R., Kerekes, E., Kocsis, B., Kocsis, M., Farkas, Á. (2021). In Vitro Antibacterial and Antibiofilm Activity of Hungarian Honeys against Respiratory Tract Bacteria. Foods, 10(7), 1632. https://doi.org/10.3390/foods10071632
14	Balázs, V.L., Nagy-Radványi, L., Bencsik-Kerekes, E., Koloh, R., Szabó, D., Kocsis, B., Kocsis, M., Farkas, Á. (2023). Antibacterial and Antibiofilm Effect of Unifloral Honeys against Bacteria Isolated from Chronic Wound Infections. Microorganisms, 11(2), 509. https://doi.org/10.3390/microorganisms11020509
15	Barbarić, A., Saftić Martinović, L., Marijanović, Z., Juretić, L., Jurič, A., Petrović, D., Šoljić, V., Gobin, I. (2025). Integrated Chemical and Biological Evaluation of Linden Honeydew Honey from Bosnia and Herzegovina: Composition and Cellular Effects. Foods, 14(10), 1668. https://doi.org/10.3390/foods14101668
16	Barvelink, M., Brok, B., Djajadiningrat-Laanen, S.C., Vernooij, J.C.M., Broens, E.M., Slenter, I.J.M. (2025). In Vitro Activity of Manuka Honey, Either Alone or in Combination With Topical Antibiotics, Against Bacteria Commonly Found in Equine Ulcerative Keratitis. Veterinary Ophthalmology, 29(2), e70111. https://doi.org/10.1111/vop.70111
17	Basson, N.J., Grobler, S.R. (2008). Antimicrobial activity of two South African honeys produced from indigenous Leucospermum cordifolium and Erica species on selected micro-organisms. BMC Complementary and Alternative Medicine, 8(1), 41. https://doi.org/10.1186/1472-6882-8-41

18	Bazaid, A.S., Aldarhami, A., Patel, M., Adnan, M., Hamdi, A., Snoussi, M., Qanash, H., Imam, M., Monjed, M.K., Khateb, A.M. (2022). The Antimicrobial Effects of Saudi Sumra Honey against Drug Resistant Pathogens: Phytochemical Analysis, Antibiofilm, Anti-Quorum Sensing, and Antioxidant Activities. Pharmaceuticals, 15(10), 1212. https://doi.org/10.3390/ph15101212
19	Bazaid, A.S., Alsolami, A., Patel, M., Khateb, A.M., Aldarhami, A., Snoussi, M., Almusheet, S.M., Qanash, H. (2023). Antibiofilm, Antimicrobial, Anti-Quorum Sensing, and Antioxidant Activities of Saudi Sidr Honey: In Vitro and Molecular Docking Studies. Pharmaceutics, 15(9), 2177. https://doi.org/10.3390/pharmaceutics15092177
20	Bereksi-Reguig, D., Allali, H., Taib, N., Aissaoui, N., Wlodarczyk-Stasiak, M., Kowalski, R. (2024). Bioactive Compounds, Antioxidant Properties, and Antimicrobial Profiling of a Range of West Algerian Honeys: In Vitro Comparative Screening Prior to Therapeutic Purpose. Foods, 13(24), 4120. https://doi.org/10.3390/foods13244120
21	Boateng, J., Diunase, K. (2015). Comparing the Antibacterial and Functional Properties of Cameroonian and Manuka Honeys for Potential Wound Healing—Have We Come Full Cycle in Dealing with Antibiotic Resistance?. Molecules, 20(9), 16068-16084. https://doi.org/10.3390/molecules200916068
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